



SCHOOL OF
**COMPUTING &
DATA SCIENCE**
The University of Hong Kong

COMP4501 / COMP4802

Capstone Projects for BEng(DS&E)

Overall Framework

2025-26

Loretta Choi

Project Coordinator

Project Requirements

Feature	COMP4501: Data science in discipline project	COMP4502: Final Year Project
Primary Focus	Application & Analysis. The project must be associated with a "domain focus" and "specific application domain". The goal is to deliver a "data-driven solution".	Development & Engineering. The project is a "full end-to-end development" of a "substantial project". Topics can be "applied technologies" or "basic research".
Kind of Problems	Domain-Specific. For example: "How can we use data to solve a problem in [finance, biology, logistics, etc.]?" "Can I build a model to predict [domain-specific outcome]?" "What insights can I find in this dataset?"	Engineering-Specific. For example: "Can I build a new tool or system for data processing?" "Can I design and implement a new algorithm or robust solution?" "Can I conduct basic research on a data science method?"
Key Project Phases	The phases emphasize the classic data science lifecycle: 1. Problem Framing 2. Data Exploration & Modeling 3. Solution Deployment & Storytelling	The phases emphasize a classic software/engineering lifecycle: 1. Project Initiation 2. Design and Prototyping 3. Implementation and Delivery
Deliverables & Assessment	100% continuous assessment based on a project plan, presentations, and written reports. The focus of these will be on the analysis, interpretation, and communication of your findings.	100% continuous assessment based on a project plan, presentations, and written reports. The focus of these will be on the design, technical implementation, and delivery of your final project.
Teamwork	2-4 students work in a project	1-4 students work in a project

COMP4501/COMP4502 Assessment

Deliverables	Supervisor	2nd Examiner
Project Plan	10	10
Final Presentation	15	15
Final Report	25	25
Project Work	50	50
Total	100	100
Weighted Total	60%	40%

Timeline

Dec 27, 2025	Short Proposal Submission
Feb 13, 2026	Project Plan Submission
Apr 27-30, 2026	Final Presentation
May 7, 2026	Final Report Submission

Reference

Course Description and Course Learning Outcomes

COMP4501 Data science in discipline project

Course Description

Students will work on a capstone project which is on data science in association with a domain focus. Students are required to identify a data-intensive problem in a specific application domain, and to implement a data-driven solution for the problem. Students will undergo a complete data science project life cycle, from problem understanding, data collection, data exploration to data modelling, analysis and interpretation, and finally deliver a data science solution.

COMP4501 Course Learning Outcomes

1. **[End-to-End Project Execution]**
Plan and execute the full lifecycle of a data science project, including problem identification, data collection, exploration, modeling, analysis, and solution delivery, mirroring real-world expectations for Data Science professionals.
2. **[Technical Implementation & Domain Integration]**
Apply data science methodologies to design and implement practical, data-driven solutions for domain-specific problems, ensuring relevance and addressing real-world needs through interdisciplinary understanding.
3. **[Communication & Documentation]**
Effectively communicate technical concepts, project progress, and results to stakeholders through verbal presentations, written reports, and live demonstrations, ensuring clarity for both technical and non-technical audiences.
4. **[Decision Making & Problem Solving]**
Navigate unforeseen challenges in data science projects, make informed decisions under constraints (e.g., data limitations, time, resources), and adapt methodologies to achieve feasible and impactful outcomes.

COMP4502 Final Year Project

Course Description

Student individuals or groups, during the final year of their studies, undertake full end-to-end development of a substantial project, taking it from initial concept through to final delivery. Topics range from applied technologies to assignments on basic research in relation to data science and engineering. In case of a team project, significant contribution is required from each member and students are assessed individually. Strict standards of quality will be enforced throughout the project development.

COMP4502 Course Learning Outcomes

1. **[End-to-End Project Execution]**
Plan and execute the full lifecycle of a data science or engineering project—from problem definition and data acquisition to deployment and maintenance—employing industry-standard method tools and methodologies.
2. **[Technical Implementation & Integration]**
Apply computational and data science methods to develop a robust solution, integrating domain knowledge and ethical considerations.
3. **[Communication & Documentation]**
Communicate technical work effectively through, presentations, and demonstrations, translating complex results for both technical and non-technical audiences.
4. **[Decision Making & Problem Solving]**
Resolve project challenges by evaluating trade-offs, iterating on solutions, and adapting to constraints (time, resources, stakeholder needs).